THE UNIVERSITY OF MELBOURNE

Student number

Semester 2 Assessment, 2023

School of Mathematics and Statistics

MAST20030 Differential Equations

Reading time: 15 minutes — Writing time: 3 hours

This exam consists of 40 pages (including this page) with 7 questions and 120 total marks

Permitted Materials

- No books, notes or other printed or handwritten material are permitted.
- No calculators are permitted.
- Any mobile phones or internet-enabled devices brought into the exam room must be **turned off** and placed on the floor under your table.

Instructions to Students

- *** Write student instructions that are not auto-loaded here, preceded by `\item`. If you selected the 'appendedmatter' option, tell students what is appended.
- ***** sample extra instruction ***
- Write your answers in the boxes provided on the exam. There is extra space you can use for answers to any question commencing on page 32. If you still do not have enough space, tick the box near the bottom of page 37 and request extra pages from an invigilator—include the question number at the top of each additional page.
- You must NOT remove this question paper, or any extra pages provided to you, at the conclusion of the examination.

Instructions to Invigilators

- Students are to write their answers on the paper. They may request extra pages if they run out of space on the exam paper.
- This exam paper contains examinable material and must be collected, together with extra pages if any, at the conclusion of the examination.
- *** Write invigilator instructions that are not auto-loaded here, preceded by `\item`.

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Question 1 (24 marks)

The system

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \boldsymbol{A} \cdot \begin{bmatrix} x \\ y \end{bmatrix} \qquad \text{where} \qquad \boldsymbol{A} = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}$$

has the general solution

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t} + \beta \left(\begin{bmatrix} 1 \\ -1 \end{bmatrix} t + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) e^{-t}$$

where $\alpha, \beta \in \mathbb{R}$.

(a) Write down the eigenvalues and linearly independent eigenvectors of $\boldsymbol{A}$.

(b) Express the system as a single 2nd order differential equation in $x(t)$.

(c) Sketch the phase portrait of the system. Include the following information:

- any straight line orbits and their directions;
- at least 4 other orbits and their directions, showing the asymptotic behaviour as $t \to \infty$ and $t \to -\infty$;
- the slopes at which the orbits cross the $x-$ and $y-$axes;
- the loci of all minima and maxima in x, and also in y, if any of these exist;
- the orbit (or orbits) where the ratio of y/x remains constant for all t and the value of this ratio;
- the type and stability of the critical point at $(0, 0)$.

Be sure to include all relevant calculations and justification for each item above.

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Question 2 (12 marks)

Consider the IVP

$$x''(t) + 2x'(t) + x(t) = 0 \qquad \text{where} \qquad x(0) = a,\ x'(0) = b$$

(a) Solve the IVP using the method of Laplace transforms.

(b) Find $\dfrac{x'(10)}{x(10)}$ given $x(0) = 1$ and $x'(0) = -1$. *You may use any results from Q1.*

(c) Find the Laplace transform of

$$\int_0^t \delta(\tau - 2)d\tau$$

where $\delta(t)$ is the dirac-delta 'function'.

(d) Use the method of Laplace transforms to find $y(t)$ given

$$\int_0^t \delta(\tau - 2)d\tau = 5\int_0^t y(\tau)d\tau + y'(t) + 2y(t), \qquad \text{where} \qquad y(0) = 0.$$

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Question 3 (22 marks)

Consider Bessel's equation of order zero

$$r^2R'' + rR' + (\omega r)^2 R = 0$$

where $\omega > 0$.

(a) By trialling a power series of the form

$$R(r) = \sum_{n=0}^{\infty} c_n r^n$$

derive a recurrence relationship for the c_n coefficients.

(b) Using your result from (a), compute c_2, c_3 and c_4.

(c) Using your results from (a) and (b), deduce an explicit expression for the c_n coefficients, and thereby write down one linearly independent series solution for $R(r)$.

(d) Using the method of reduction of order, we will now find a second linearly independent solution to Bessel's equation of order zero.

(i) Given

$$R(r) = u(r)J_0(\omega r),$$

where $J_0(\omega r)$ is a solution to Bessel's equation of order zero, show that

$$u'' + \left(\frac{2J_0'}{J_0} + \frac{1}{r}\right)u' = 0.$$

(ii) Using the ODE for u in part (i), solve for u. Express u in terms of an integral involving $J_0(\omega r)$. Do not attempt to evaluate this integral.

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Question 4 (25 marks)

(a) Consider the function

$$f(x) = \frac{1}{2}, \quad \text{for} \quad 0 < x < 1, \quad f(0) = f(1) = 0, \quad f(-x) = -f(x), \quad f(x+2) = f(x).$$

(i) Is the function $f(x)$ even, odd or neither?

(ii) Calculate the Fourier series of $f(x)$.
Only calculate the non-zero coefficients based on your answer in (i).

(b) Consider the function

$$g(x) = x - \frac{\pi}{2} \qquad \text{for} \qquad x \in [0, \pi], \qquad g(-x) = g(x), \qquad g(x + 2\pi) = g(x).$$

The Fourier series of $g(x)$, where $L = \pi$, is

$$s(x) = -\frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2m-1)^2} \cos\big([2m-1]x\big).$$

(i) Sketch $g(x)$ for $x \in [-3\pi, 3\pi]$.

(ii) Use the pointwise convergence theorem, and $s(x)$, to evaluate the sum

$$\sum_{m=1}^{\infty} \frac{1}{(2m-1)^2}.$$

(iii) Does the Fourier series of $g(x)$ exhibit Gibbs phenomenon? If so, what is the error of the truncated Fourier series approximation at this point?

(iv) Does the Fourier series of $g(x)$ converge uniformly? If so, prove this using the Weierstrass test for uniform convergence.

(v) Find an antiderivative of $s(x)$ using

$$S(x) = \int_0^x s(\xi)d\xi$$

and term by term integration.

(vi) Is the term-by-term integration in (e) justified? Explain your answer in one sentence.

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Question 5 (19 marks)

Consider the axisymmetric vibrations of a circular drum membrane of radius a, which are governed by

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r}, \qquad 0 \leq r < a, \qquad t > 0.$$

The membrane is held fixed at the boundary to maintain $u(a,t) = 0$, and is released from rest with an initial displacement profile $u(r,0) \neq 0$. Find the general series solution for the displacement $u(r,t)$ in a sequence of steps as follows.

(a) Using $u(r,t) = R(r)T(t)$, derive the equations governing both $R(r)$ and $T(t)$. *Do not apply any boundary or initial conditions at this step.*

(b) Write down the three cases for the constant of separation. State which of these leads to *oscillatory vibrations in time.*

(c) Here consider only the case that leads to oscillatory vibrations.

(i) Solve the resulting BVP for $R(r)$. *Use any relevant result from Q3. Do not repeat calculations you have already done.*

(ii) Solve the ODE for $T(t)$.

(iii) Find the general series solution to the initial boundary value problem. *Do not attempt to find a formula for the remaining series coefficients.*

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Question 6 (11 marks)

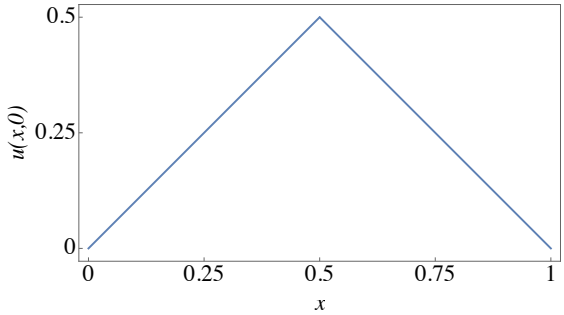
(a) Consider the heat diffusion process along a rod of unit length, as governed by

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

subject to the Neumann (reflecting) boundary conditions

$$\partial_x u(0,t) = \partial_x u(1,t) = 0,$$

and the following initial condition $u(x,0)$ for $x \in [0,1]$



Not covered in 2025

(i) Find the total heat in the system at $t = 0$. What is the steady state temperature distribution along the rod? Justify your answer in one sentence. *Do not solve the full separation of variables problem.*

(ii) Sketch $u(x,0)$, $u(x,t_1)$, $u(x,t_2)$ and $u(x,\infty)$, $0 < t_1 < t_2 < \infty$, on one set of axes.

(b) Consider the heat and wave equations:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, \qquad \text{and} \qquad \frac{\partial^2 v}{\partial t^2} = \frac{\partial^2 v}{\partial x^2}$$

which are both subject to homogeneous Neumann (reflecting) boundary conditions at $x = 0$ and $x = 1$, and the same initial condition

$$u(x, 0) = v(x, 0) = H(x - 1/2)$$

where $H(x)$ is the Heaviside function. In 1-2 sentences, discuss how the discontinuity in the initial condition propagates forwards in time for both the heat and wave equations. *Do not perform any calculations here.*

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Question 7 (7 marks)

Consider the ODE

$$\frac{d^5y}{dx^5} + xy = 0$$

on the open domain $x \in \mathbb{R}$.

Use the method of Fourier transforms to find the most general solution to the ODE that has a Fourier transform. You may leave your final answer as a definite integral.

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End of Exam — Total Available Marks = 120

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You must tick this box if you have used extra pages ☐

MAST20030 Differential Equations Formulae Sheet

1) **Laplace Transforms**

1.	$f(t)$	$(\mathcal{L}f)(s) = F(s) = \int_0^\infty f(t)e^{-st}\,dt$	(Definition of Transform)
2.	1	$\frac{1}{s}$	
3.	t^n	$\frac{n!}{s^{n+1}}$	
4.	e^{at}	$\frac{1}{s-a}$	
5.	$\sin(at)$	$\frac{a}{s^2+a^2}$	
6.	$\cos(at)$	$\frac{s}{s^2+a^2}$	
7.	$\sinh(at)$	$\frac{a}{s^2-a^2}$	
8.	$\cosh(at)$	$\frac{s}{s^2-a^2}$	
9.	$\delta(t-a)$	e^{-as}	$(a \geq 0)$
10.	$f'(t)$	$sF(s) - f(0)$	
11.	$f''(t)$	$s^2F(s) - sf(0) - f'(0)$	
12.	$f^{(n)}(t)$	$s^nF(s) - \sum_{k=0}^{n-1} s^{n-1-k} f^{(k)}(0)$	
13.	$\int_0^t f(\tau)\,d\tau$	$\frac{F(s)}{s}$	
14.	$e^{-at}f(t)$	$F(s+a)$	(s-Shifting Theorem)
15.	$f(t-a)\,u(t-a)$	$e^{-as}F(s)$	($a > 0$, t-Shifting Theorem) $u(t)$ is the Heaviside step function
16.	$\int_0^t f(\tau)\,g(t-\tau)\,d\tau$	$F(s)\,G(s)$	(Convolution)

2) **Standard Limits**

(i) $\lim_{n\to\infty} \frac{1}{n^p} = 0 \quad (p > 0)$

(ii) $\lim_{n\to\infty} r^n = 0 \quad (|r| < 1)$

(iii) $\lim_{n\to\infty} a^{1/n} = 1 \quad (a > 0)$

(iv) $\lim_{n\to\infty} n^{1/n} = 1$

(v) $\lim_{n\to\infty} \frac{a^n}{n!} = 0 \quad (\text{all } a)$

(vi) $\lim_{n\to\infty} \frac{\log_e n}{n^p} = 0 \quad (p > 0)$

(vii) $\lim_{n\to\infty} \left(1 + \frac{a}{n}\right)^n = \mathrm{e}^a \quad (\text{all } a)$

(viii) $\lim_{n\to\infty} \frac{n^p}{a^n} = 0 \quad (\text{all } p, a > 1)$

3) **The Generalised Harmonic Series (p-Series)**

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \quad \text{is} \quad \begin{cases} \text{convergent} & \text{if } p > 1 \\ \text{divergent} & \text{if } p \leq 1 \end{cases}$$

4) **Geometric Series**

$$\sum_{n=0}^{\infty} ar^n \quad \text{is} \quad \begin{cases} \text{convergent} & \text{if } |r| < 1 \\ \text{divergent} & \text{if } |r| \geq 1 \end{cases}$$

5) **Fourier Series Formulae**

$$\begin{aligned} f(x) &= \frac{1}{2}a_0 + \sum_{n=1}^{\infty} (a_n \cos k_n x + b_n \sin k_n x), \quad k_n = \frac{n\pi}{L} \\ a_0 &= \frac{1}{L}\int_{-L}^{L} f(x)\, dx \\ a_n &= \frac{1}{L}\int_{-L}^{L} f(x) \cos k_n x\, dx \\ b_n &= \frac{1}{L}\int_{-L}^{L} f(x) \sin k_n x\, dx \end{aligned}$$

6) **Fourier Transforms Formulae**

$$\widehat{f}(k) = (\mathscr{F} f)(k) = \int_{-\infty}^{\infty} f(x)\mathrm{e}^{-ikx}\, dx$$

$$f(x) = (\mathscr{F}^{-1}\widehat{f})(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widehat{f}(k)\mathrm{e}^{ikx}\, dk$$

$$\mathscr{F}\left\{\frac{\mathrm{d}^n f}{\mathrm{d}x^n}\right\} = (ik)^n \widehat{f}(k), \qquad \mathscr{F}\left\{x^n f(x)\right\} = i^n \frac{\mathrm{d}^n \widehat{f}}{\mathrm{d}k^n}$$

7) **Trigonometric and Hyperbolic Formulae**

$$\cos^2 x + \sin^2 x = 1 \qquad \cosh^2 x - \sinh^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x \qquad 1 - \tanh^2 x = \operatorname{sech}^2 x$$

$$\cot^2 x + 1 = \operatorname{cosec}^2 x \qquad \coth^2 x - 1 = \operatorname{cosech}^2 x$$

$$\sin(x+y) = \sin x \cos y + \cos x \sin y \qquad \sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y$$

$$\cos(x+y) = \cos x \cos y - \sin x \sin y \qquad \cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y$$

$$\sin x \sin y = \frac{1}{2}\left[\cos(x-y) - \cos(x+y)\right] \qquad \sinh x \sinh y = \frac{1}{2}\left[\cosh(x+y) - \cosh(x-y)\right]$$

$$\cos x \cos y = \frac{1}{2}\left[\cos(x-y) + \cos(x+y)\right] \qquad \cosh x \cosh y = \frac{1}{2}\left[\cosh(x+y) + \cosh(x-y)\right]$$

$$\sin x \cos y = \frac{1}{2}\left[\sin(x-y) + \sin(x+y)\right] \qquad \sinh x \cosh y = \frac{1}{2}\left[\sinh(x+y) + \sinh(x-y)\right]$$

$$\sin x = \frac{i}{2}\left(\mathrm{e}^{-ix} - \mathrm{e}^{ix}\right) \qquad \sinh x = \frac{1}{2}\left(\mathrm{e}^{x} - \mathrm{e}^{-x}\right)$$

$$\cos x = \frac{1}{2}\left(\mathrm{e}^{-ix} + \mathrm{e}^{ix}\right) \qquad \cosh x = \frac{1}{2}\left(\mathrm{e}^{x} + \mathrm{e}^{-x}\right)$$